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Division: A2

Subject : Programming with java

Assignment 2: Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Program:

```
class Student {  
    int id;  
    String name;  
    double marks;  
    Student() {  
        id = 0;  
        name = "Not Assigned";  
        marks = 0.0;  
    }  
    Student(int id, String name, double marks) {  
        this.id = id;  
        this.name = name;  
        this.marks = marks;  
    }  
    Student(Student s) {  
        this.id = s.id;  
        this.name = s.name;  
        this.marks = s.marks;  
    }  
    void display() {
```

```

        System.out.println("ID   : " + id);
        System.out.println("Name  : " + name);
        System.out.println("Marks : " + marks);
        System.out.println();
    }
}

public class ConstructorDemo {
    public static void main(String[] args) {
        Student s1 = new Student();
        Student s2 = new Student(101, "Rahul", 89.5);
        Student s3 = new Student(s2);
        System.out.println("Student created using Default Constructor:");
        s1.display();
        System.out.println("Student created using Parameterized Constructor:");
        s2.display();
        System.out.println("Student created using Copy Constructor:");
        s3.display();
    }
}

```

Question 1: Create a Student class using default and parameterized constructors to initialize student name and roll number.

```

class Student {
    String name;
    int rollNo;
    Student() {
        name = "atharv";
        rollNo = 053;
    }
}

```

```
}
```

```
Student(String n, int r) {
```

```
    name = n;
```

```
    rollNo = r;
```

```
}
```

```
void display() {
```

```
    System.out.println(name + " " + rollNo);
```

```
}
```

```
public static void main(String[] args) {
```

```
    Student s1 = new Student();
```

```
    Student s2 = new Student("Rahul", 101);
```

```
    s1.display();
```

```
    s2.display();
```

```
}
```

```
}
```

Question 2: Develop a mobile inventory system using different constructors to initialize mobile details and duplicate object records

```
class Mobile {
```

```
    String brand;
```

```
    int price;
```

```
    Mobile() {
```

```
        brand = "Samsung";
```

```
        price = 15000;    }
```

```
    Mobile(String b, int p) {
```

```
        brand = b;
```

```
        price = p;
```

```
}  
  
Mobile(Mobile m) {  
    brand = m.brand;  
    price = m.price;  
}  
  
void display() {  
    System.out.println(brand + " " + price);  
}  
  
public static void main(String[] args) {  
    Mobile m1 = new Mobile();           // Default  
    Mobile m2 = new Mobile("iPhone", 80000); // Parameterized  
    Mobile m3 = new Mobile(m2);         // Copy  
    m1.display();  
    m2.display();  
    m3.display();  
}  
}
```